P1 May 2010 TZ1

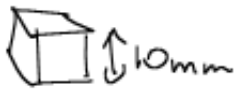
1.

1. size of nucleus ~ 10^{-15}m

$v = \frac{d}{t} \Rightarrow t = \frac{d}{v} = \frac{10^{-15}}{3\times10^{8}}$

$= 1/3 \times 10^{-23}$ A

2.

2. 

% uncertainty = $2/10 \times 100 = 20\%$

% uncertainty in volume = $3 \times 20 = 60\%$

Volume = $10\times10\times10 = 1000\,mm^3$

unc. = $\frac{60}{100} \times 1000 = 600\,mm^3$

D

3.

3. 

gradient = accel
area = displacement
and of course the speed can be determined directly but not velocity

D

4.

4.
A, $F \propto m$ only if acceleration is constant
B, rate of change of momt. is proportional to the force
C, momentum is not $\propto$ to force its rate of change
D, The acceleration is proportional to the resultant force not "A force".
B

5.

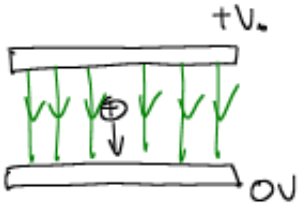
5.
Work = force × distance moved in direction of force
B

6.

6.


accel. is always towards the centre. C

7.

7.


+ve charge would accel downwards
- field uniform
C

8.

8.

Potential is WD per unit mass moving a small test mass from ∞ to the point.

A

The mass doesn't really have to be small. In fact in astronomical terms a 1kg mass is small.

9.

9.

When a body is thrown away from the earth $\Delta KE = \Delta PE$

$\frac{1}{2}mv^2 = \frac{GMm}{r}$ $v = \sqrt{\frac{2GM}{r}}$

so depends on radius & mass

10.

10.

There will be no transfer of energy since ice and water are at 0°C

→ no ice melts + no temp. change

C

11.

11.

A mole is avogadro's number of atoms ⇒ same number as in 12g of ^{12}C

B

12.

12.



When the piston moves it hits the atoms like a tennis racket hitting a ball.

A

13.

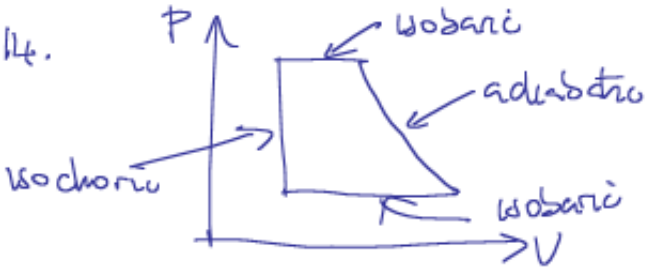
13.

Gases behave ideally when the atoms are far apart ⇒ Low P.

B

14.

14.



can't really say that II is adiabatic but its the only possibility.

D

15.

15.

The velocity is sinusoidal ⇒ SHM

The total energy of a body moving with SHM = PE + KE = constant

C

16.

16.

Resonance occurs when the frequency of the driving force = natural frequency of the system.

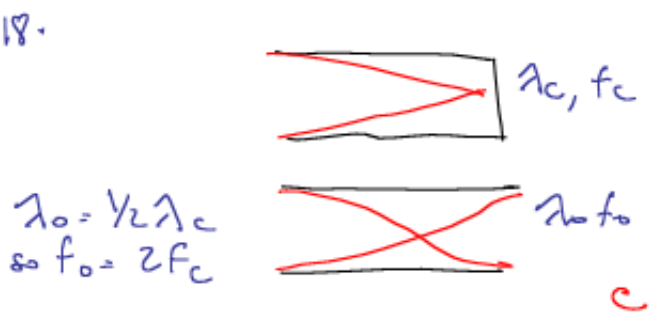
C

17.

17. visible ~ 600nm
6×10^{-7} m
so 4×10^{-7} is nearest.

B

18.



C

19.

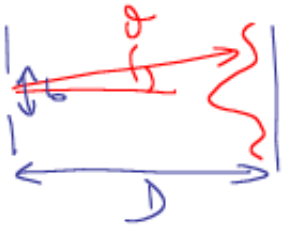
19.

When ultrasound waves reflect off moving blood cells the frequency is changed – Doppler effect.

D

20.

20.



$b \sin\theta = \lambda \quad \frac{\lambda}{b} = \sin\theta = \frac{y}{D}$

$y = \frac{\lambda D}{b} \Rightarrow$ increased if D increased b decreased

A

21.

21.



Just resolved when the 1st min^{n} of one coincides with the max of the other

D

22.

22.

An optically active substance changes the direction of polarisation

C

23.

23.

EMF – the amount of chemical → electrical energy per unit charge $\frac{E}{Q}$ but this isn't an option

however $P = E/t$ & $I = Q/t$ so $P/I = E/Q$

A

24.

24.

On earth weight of 1kg = 10N

$$10N = \frac{GM_E \times 1}{r_E^2} \Rightarrow M_E = \frac{10r_E^2}{G}$$

On mars weight of 1kg = 4N

$$4N = \frac{GM_m \times 1}{r_m^2} \quad M_m = \frac{4r_m^2}{G}$$

$$\frac{M_m}{M_E} = \frac{4r_m^2}{G} \Big/ \frac{10r_E^2}{G} = \frac{4r_m^2}{10r_E^2}$$

but $r_m = \frac{r_E}{2}$

$$\Rightarrow \frac{M_m}{M_E} = \frac{4\left(\frac{r_E}{2}\right)^2}{10r_E^2} = \frac{4/4}{10}$$

$$= 1/10 = 0.1$$

A

maybe a quicker way but it makes up for all the easy questions up to here

25.

25.

The induced current → heat.
this energy comes from the mechanical energy of the swinging plate
⇒ amplitude will decrease

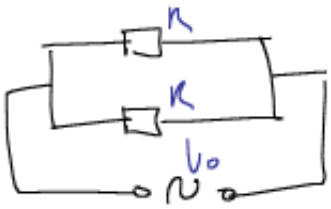
As the plate swings back and forth its direction changes ⇒ direction of current will change

⇒ both change

B

26.

26.



Total resistance = $\frac{1}{2}R$

Power = $\frac{V_{rms}^2}{R}$ $V_{rms} = \frac{V_0}{\sqrt{2}}$

$\Rightarrow V_{rms}^2 = \frac{V_0^2}{2}$ Power = $\frac{V_0^2/2}{R/2}$

$= \frac{V_0^2}{R}$

B

27.

27.

Coulomb interaction is between charges so not the neutron

strong interaction is between all nucleons

C

28.

28.

$KE_{max} = hf - \phi$ ← work function (energy to set e⁻ out)

↑ energy from a photon

A

29.

29.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

if accelerated by PD V

$$KE = Ve = \tfrac{1}{2}mv^2$$

$$\Rightarrow v = \sqrt{\frac{2Ve}{m}} \quad \therefore p = mv = m\sqrt{\frac{2e}{m}}$$

$$= \sqrt{\frac{2Vem^2}{m}} = \sqrt{2Vem}$$

$$\lambda = \frac{h}{\sqrt{2Vem}}$$

A

30.

30.

Schrodinger stated that the probability of finding an electron at any point around the nucleus is defined by the square of some wavefunction ψ.

D

31.

31

The mass spectrometer uses E & B fields to deflect charged particles so they must be ions

B

32.

32.

Since the neutrino is not an antineutrino the particle is β^+

$$\Rightarrow p \rightarrow n \quad \therefore \quad z = 18$$

B

33.

33. 16 8 4 2 1
1 1 0 1 0 $16+8+2=26$
↑
MSB
C

34.

34.
$Q = 10^4 \times 1.6\times10^{-19} = 1.6\times10^{-15}C$

$C=\frac{Q}{V} \Rightarrow V=\frac{Q}{C}=\frac{1.6\times10^{-15}}{3.2\times10^{-12}}=\frac{1}{2}\times10^{-3}$

$=5\times10^{-4}V$

B

35.

35.
quantum efficiency is high if the number of e^- is close to the number of photon

$=\frac{\text{no. of } e^-}{\text{no. of photon}}$ B

36.

36.
According to the 2nd Law all energy becomes degraded. This means it can't be used to do work.
D

37.

37. $\varepsilon = \frac{\text{Power out}}{\text{Power in}} = \frac{P}{\eta \times Q}$

Q = energy density

$Q = \frac{P}{\eta \varepsilon}$ D

38.

38.
The moderator slows down the n^o so they are absorbed more easily.

The control rods slow the reaction by removing n^o

A

39.

39.
photo voltaic is one of those electrical solar panels
solar → electric.

The heating panel is one of those heating units on the top of houses
solar → thermal

A

Duh ...

40.

40.
Albedo is to do with reflected light
glaciers reflect more than ocean
→ higher albedo

D

All materials on this website are for the exclusive use of teachers and students at subscribing schools for the period of their subscription. Any unauthorised copying or posting of materials on other websites is an infringement of our copyright and could result in your account being blocked and legal action being taken against you.